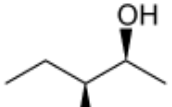
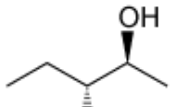
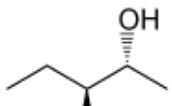
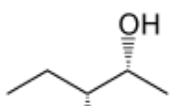


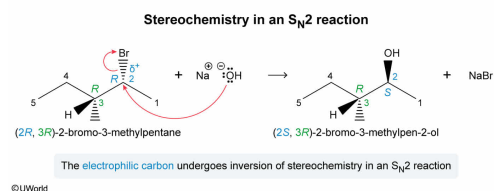
If (2*R*,3*R*)-2-bromo-3-methylpentane undergoes an S_N2 reaction with OH⁻, which of the following compounds is the reaction product?

- ☐ A.  [16%]
- ✓ ☒ B.  [52%]
- ☐ C.  [19%]
- ☐ D.  [11%]

Correct

52% answered correctly

Explanation:



The stereochemistry of a **chiral center** is designated as **R** or **S** based on the priority and circular arrangement of its substituents:

- The highest **priority** substituent (ie, priority 1) is the substituent with the greatest atomic number. If two substituent atoms are the same, the atomic numbers of the next attached atoms are considered.
- If the lowest priority group is positioned to point into the plane (dashed), a clockwise arrangement of groups 1, 2, and 3 gives an *R* configuration whereas a counterclockwise arrangement gives an *S* configuration.
- If the lowest priority group is pointing out of the plane (wedge), a clockwise arrangement of groups 1, 2, and 3 gives an *S* configuration whereas a counterclockwise arrangement gives an *R* configuration.

An **S_N2 reaction** is a substitution reaction in which a nucleophile attacks an electrophile and displaces a leaving group in one step. If the **electrophile** is a chiral center, its **stereochemistry** will be **inverted** by the reaction. Therefore, an electrophile that has an *R* configuration before substitution adopts an *S* configuration after the reaction and vice versa, if the nucleophile has the same priority ranking as the leaving group.

In this S_N2 reaction, the OH⁻ (nucleophile) attacks carbon 2 in (2*R*,3*R*)-2-bromo-3-methylpentane while

bromide (leaving group) leaves. Because the electrophile (carbon 2) is chiral, its configuration is inverted from *R* to *S*. Carbon 3 is also chiral but retains the *R* configuration because it does not participate in the reaction. The structure in Choice B correctly shows carbon 2 in the *S* configuration and carbon 3 in the *R* configuration.

(Choices A, C, and D) The configuration of carbon 2 should be *S* (wedge in this case) because it is the electrophile, so its configuration inverts in the S_N2 reaction. The configuration of carbon 3 should remain *R* (dashed in this case) because it does not participate in the S_N2 reaction.

Educational objective:

An S_N2 reaction is a concerted substitution reaction where a nucleophile forms a bond with an electrophile while a leaving group is displaced. If the electrophile is a chiral center, its stereochemistry will be inverted in the product.

Time spent:0 sec

Subject:
Organic Chemistry

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System:
Introduction to Organic Chemistry